A Formally Verified Multi-Start Steffensen Scheme for the Equation $z^p = x$ in $\mathbb{C}$

Lean 4 / Mathlib v4.7.0 development, with a companion spectral framework on the derivative values at cyclotomic roots

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Abstract

We present a Lean 4 / Mathlib v4.7.0 development of a multi-start variant of the Pandrosion-Steffensen iteration for solving $z^p = x$ with $x \in \mathbb{C}^*$ and $p \geq 1$. The scheme combines the rational iteration $h_{p,x}(z) = 1 - (x-1)/(x\,S_p(z))$ (where $S_p(z) = \sum_{k=0}^{p-1} z^k$), the Aitken-Steffensen Δ^2 accelerator $\sigma = \mathrm{Aitken}(h)$, and a perturbative seed $\mathrm{seed}_{\alpha,\varepsilon}(z) = \alpha + \varepsilon(z - \alpha)$ whose definition depends on the user-chosen cyclotomic target α.

The paper reports three kinds of content, explicitly separated: *(a) results that are new in this formalization*, concerning the universal convergence of the multi-start scheme with an explicit closed-form iteration count, the Borel-measurability and a.e. continuity of the resulting solution map, and a set of elementary algebraic identities for the finite series $\zeta_P(s) := \sum_{k=1}^{p} P'(\alpha_k)^{-s}$; *(b) Lean specializations of classical theorems (Banach 1922, Gauss 1799, Aitken 1926, Schwarz 1890, Cardano 1545, Vieta 1591, Kronecker 1857, Galois 1832, Aberth–Ehrlich 1967/1973) to the Pandrosion setting these are not new mathematics, only new formalizations*; *(c) two short structural observations: a compliance interface for the Kung–Traub upper bound for $c \geq 3$, and a statement-level recording of McMullen 1987. Two other classical themes (Lehmer's Mahler-measure conjecture, Smale's 17th problem) that earlier drafts discussed have been removed from scope, because the Pandrosion setting contributes nothing to either.*

The Lean development comprises 121 modules with zero `sorry` and an audited dependency closure reducing to {`propext`, `Classical.choice`, `Quot.sound`}, verifiable per-theorem via `#print axioms` (see Section 10). The main contribution of the corpus is the formalization itself and the multi-start convergence analysis of (a), not any of the engagements in (c); readers primarily interested in the mathematics should focus on Sections 2–6.

Contents

1 Introduction and scope

1.1 Historical starting point: Pandrosion's two sequences for $\sqrt[3]{2}$

Before stating the general iterator, we recall the two sequences that Pappus (*Collectio*, Book III) attributes to Pandrosion of Alexandria for approximating $\sqrt[3]{2}$. Given an initial parameter $u_0 \in (0,4)$, the construction produces the pair

$$u_{n+1} \;=\; \frac{u_n}{2 \;-\; 2\left(\dfrac{4-u_n}{4}\right)^3}, \qquad v_n \;=\; 2\left(\frac{4-u_n}{4}\right)^2, \qquad u_0 \;=\; 0.5. \tag{1}$$

The sequence (u_n) is the internal parameter (a length in the Thales diagram); the sequence (v_n) is the *readout*, converging to $\sqrt[3]{2}$.

Normalization. The substitution $s_n := (4-u_n)/4 \in (0,1)$ eliminates all constants:

$$u_n \;=\; 4(1-s_n), \qquad v_n \;=\; 2s_n^2, \qquad u_{n+1} \;=\; \frac{4(1-s_n)}{2(1-s_n^3)} \;=\; \frac{2}{1+s_n+s_n^2},$$

which turns (1) into the single rational iteration

$$s_{n+1} \;=\; 1-\frac{u_{n+1}}{4} \;=\; \frac{2s_n^2+2s_n+1}{2s_n^2+2s_n+2}. \tag{2}$$

The fixed point of (2) is $s^* = 2^{-1/3}$, giving $v^* = 2\cdot 2^{-2/3} = 2^{1/3} = \sqrt[3]{2}$, and the local convergence is *linear* with ratio

$$\lambda_{3,2} \;=\; \frac{(\sqrt[3]{2}-1)(\sqrt[3]{2}+2)}{\sqrt[3]{4}+\sqrt[3]{2}+1} \;\approx\; 0.2202.$$

Recognition of the general form. Writing $x=2$, $p=3$, and $S_3(s) = 1+s+s^2$, the iteration (2) rearranges to

$$s_{n+1} \;=\; 1-\frac{1}{2(1+s_n+s_n^2)} \;=\; 1-\left.\frac{x-1}{x\cdot S_p(s_n)}\right|_{p=3,\,x=2}.$$

This is exactly the iterator $h_{p,x}$ introduced below, at $(p,x) = (3,2)$. The fixed-point equation $s^* = 1-(x-1)/(x\,S_p(s^*))$ rearranges to $x\cdot(s^*)^p = 1$, i.e. $s^* = x^{-1/p}$, and the readout $v^* = x\cdot(s^*)^{p-1} = x^{1/p}$ recovers the target p-th root. The historical construction (1) is therefore the $(p,x) = (3,2)$ instance of a one-parameter family defined in (3).

1.2 The general iterator $h_{p,x}$

On $\mathbb{C}^*$ with $p \in \mathbb{N}_{\geq 1}$, the Pandrosion iterator is

$$h_{p,x}(z) \;:=\; 1-\frac{x-1}{x\cdot S_p(z)}, \qquad S_p(z) \;:=\; \sum_{k=0}^{p-1} z^k, \tag{3}$$

whose fixed points are exactly the p-th roots of $1/x$: $h_{p,x}(s) = s \iff s^p = 1/x$. The historical case (1) is the instance $(p,x) = (3,2)$. Two constructions operate on top of $h_{p,x}$:

- an *algebraic acceleration*, the Aitken Δ^2 transform $\sigma_{p,x} := \mathrm{Aitken}(h_{p,x})$, which upgrades linear to quadratic convergence without computing derivatives (Section 2);
- a *seeded multi-start* in $\mathbb{C}$, placing z_0 inside a controlled disk around a user-chosen cyclotomic anchor before iteration begins (Section 4).

Both are used below. We first analyze $h_{p,x}$ on the real line (Section 2), where the closed-form contraction ratio and preconditionings are cleanest, then extend to the complex case via multi-start.

1.3 Contributions, separated by kind

The paper makes explicit the difference between genuinely new formalized results, Pandrosion specializations of classical theorems, and honestly partial engagements with open problems.

Kind	**What is in the corpus**	**Where**
New real-line results for $h_{p,x}$	Closed-form linear contraction ratio $\lambda_{p,x}$ (called the "convergence signature" for presentation; this is of course just $h'_{p,x}(s^*)$), optimal derivative-free starting point $s_0^{\mathrm{opt}} = h_{p,x}(1)$, scaling preconditioning $x^{1/p} = A^{1/p}(x/A)^{1/p}$ exploiting monotonicity of λ, Aitken Δ^2 upgrade to quadratic convergence with explicit constants.	Section 2.
New multi-start results in $\mathbb{C}$	Universal multi-start convergence, explicit closed-form iteration count, Borel-measurability and a.e. continuity of the multi-start solution map.	Sections 4, 5.
Elementary algebraic identities for ζ_P	Vanishing identity $\zeta_P(1) = 0$, $(p-1)$ higher-vanishing relations + normalization, closed-form product $\prod_k P'(\alpha_k) = (-1)^{p-1}p^p/x^{p-1}$.	Section 6. All of these reduce to the geometric-sum identity on roots of unity; we use a "finite Riemann-style" vocabulary only as a presentation device and claim no number-theoretic content.
Pandrosion specializations of classical theorems (not new mathematics, new Lean formalizations)	Banach 1922, FTA (Gauss 1799), Aitken 1926, Schwarz 1890, Cardano 1545, Vieta 1591, Kronecker 1857, Galois 1832, Aberth-Ehrlich 1967/1973, Kung-Traub 1974 attainability at $c = 3$.	Section 7.
Two short structural observations	Kung–Traub upper bound for $c \geq 3$ (compliance interface, not a proof); statement-level recording of McMullen 1987 without a formal proof.	Section 8. Lehmer's conjecture and Smale's 17th problem are deliberately outside scope (earlier drafts discussed them; they have been removed).

The paper does *not* claim to solve any of the four classical open problems in Section 8. Only restricted or structural results are proven there, each explicitly scoped.

1.4 Related work

Rational iterations for p-th roots are classical. Newton's method applied to $f(u) = u^p - x$, Halley's method, and the Schröder (1870) and König (1884) families supply iterations of arbitrary order with explicit asymptotic constants; Traub's *Iterative Methods for the Solution of Equations* (1964) is the canonical systematic treatment of fixed-point iterations and their acceleration, and Ortega–Rheinboldt (1970) treats the multidimensional analogue. Aitken–Steffensen Δ^2 acceleration is standard textbook material [**?**]. The cyclotomic identities in Section 6 are elementary consequences of the geometric-sum vanishing $\sum_k \omega^{ks} = 0$ on roots of unity, which is used in every undergraduate treatment of cyclotomic polynomials.

What is specific to this paper is: (i) the rational iteration $h_{p,x}$ arising from the Pandrosion construction, which is *not* the Newton iteration of any polynomial and has order one rather than two; (ii) the closed-form ratio $\lambda_{p,x}$ and the preconditioning built on its monotonicity; (iii) the multi-start seed of §4.1 together with the measurability/a.e.-continuity analysis of the induced solution map; (iv) the Lean 4 formalization itself, which is the main deliverable. We do not claim the iteration itself, or the ancillary identities, are novel mathematics; the contribution is to assemble them in a verified formal framework and to record the specific Pandrosion parameters.

1.5 On naming and scope

A remark on terminology. We define $\zeta_P(s) := \sum_{k=1}^{p} P'(\alpha_k)^{-s}$ as a finite series of p complex terms. All the identities we prove for it (including $\zeta_P(1) = 0$ and a closed-form for $\prod_k P'(\alpha_k)$) reduce to the geometric-sum identity on roots of unity and are elementary. We occasionally use a vocabulary reminiscent of the Riemann zeta function ("integer zeros of ζ_P") because the shape of the identities mirrors it, but we do *not* claim any number-theoretic content and the analogy should be read as typographic, not mathematical. In particular, nothing in this paper bears on the Riemann Hypothesis or on any transcendental question.

The phrase "Niveau 5 conjecture" is our own internal name for an a.e.-convergence statement on the plain Pandrosion iteration that we use as a target of Theorem 4.5; it does not appear under this label in the classical literature.

1.6 Reading order

Section 2 develops the real-line theory of $h_{p,x}$: closed-form contraction ratio, optimal starting point, scaling preconditioning, and Aitken Δ^2 acceleration. Section 3 collects the cyclotomic identities for the complex setting. Sections 4–5 contain the multi-start results in $\mathbb{C}$ (universal convergence, iteration count, measurability and a.e. continuity). Section 6 collects the finite-sum identities for ζ_P. Section 7 gathers the Pandrosion specializations of classical theorems. Section 8 records the restricted / partial engagements with the four open problems. Sections 9 and 10 describe the Lean corpus structure and how to reproduce everything. A reader interested only in the new mathematical content can restrict attention to Sections 2–5 and 6.

2 The Pandrosion iteration on the real line

This section studies $h_{p,x}$ on $\mathbb{R}_{>0}$, where every quantity admits a closed form. The results of independent interest are three: a rational expression for the linear contraction ratio $\lambda_{p,x}$ depending only on the target (§2.1), an optimal derivative-free starting point (§2.2), and a scaling preconditioning that collapses the iteration count for large x (§2.3). Sections 2.4 and 2.5 then show that Aitken's Δ^2 turns $h_{p,x}$ into the quadratically convergent $\sigma_{p,x}$ with a fully explicit local bound; this is the tool that drives the complex analysis of Section 4.

Throughout, $x > 1$ and $p \geq 2$; $\alpha := x^{1/p} > 1$ is the target p-th root and $s^* := 1/\alpha = x^{-1/p}$ is the unique fixed point of $h_{p,x}$ in $(0, 1)$. The readout is $v_n := x \cdot s_n^{p-1}$, converging to $v^* = \alpha$.

2.1 Closed-form contraction ratio (convergence signature)

Theorem 2.1 (Closed-form contraction ratio)**.** *For $x > 1$, $p \geq 2$, the iteration $s_{n+1} = h_{p,x}(s_n)$ is linearly convergent at s^* with ratio*

$$\lambda_{p,x} \;=\; h'_{p,x}(s^*) \;=\; \frac{(\alpha - 1)\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k}, \qquad \alpha = x^{1/p}. \tag{4}$$

In particular, $\lambda_{p,x} \in (0, 1)$ is a rational function of the target α alone.

Direct computation: $h'_{p,x}(s) = (x-1)S'_p(s)/(xS_p(s)^2)$, evaluated at $s^* = 1/\alpha$ and simplified using $x = \alpha^p$ and $xS_p(s^*) = \alpha/(\alpha - 1)$.

Why this matters. The number of iterations to reach precision δ is governed by

$$n_\delta \;\approx\; \left\lceil \frac{\log(\varepsilon_0/\delta)}{\log(1/\lambda_{p,x})} \right\rceil, \qquad \varepsilon_0 := |v_0 - \alpha|. \tag{5}$$

Two levers for reducing n_δ are therefore available: reducing ε_0 (one-step preconditioning of the argument, §2.2) and reducing $\lambda_{p,x}$ (scaling of the target, §2.3). We call $\lambda_{p,x}$ the *convergence signature* of the method: it encodes the method-target interaction independently of the starting point.

Monotonicity and limits. A further computation (carried out in Lean in the corpus) shows that $\lambda_{p,x}$ is strictly increasing in both p and x on $[1, \infty)$. The limits are

$$\lim_{x\to\infty} \lambda_{p,x} \;=\; 1^- \quad (\text{for fixed } p), \qquad \lim_{p\to\infty} \lambda_{p,x} \;=\; 1 - \frac{\ln x}{x-1} \;\in\; (0,1) \quad (\text{for fixed } x > 1).$$

In particular, for fixed x the ratio does *not* tend to 1 as p grows it saturates at $1 - \ln(x)/(x-1)$. The scaling preconditioning of §2.3 is built directly on the monotonicity in x.

2.2 Optimal starting point: one-step preconditioning

Proposition 2.2 (Optimal α-free starting point)**.** *Among starting points $s_0 \in (0,1)$ computable without knowledge of α, the choice*

$$s_0^{\text{opt}} \;:=\; h_{p,x}(1) \;=\; 1 - \frac{x-1}{x\,p} \tag{6}$$

minimises the initial residual $|s_0 - s^*|$ to leading order.

Since $h_{p,x}$ is the method's own one-step update and s^* is its fixed point, applying $h_{p,x}$ to the naive boundary value $s = 1$ gives the best single-step estimate of s^* that does not require α. Taylor expansion at $s = 1$ gives $s^* = 1 - (\ln x)/p + O(1/p^2)$, so $|s_0^{\text{opt}} - s^*| = |(x-1)/(xp) - (\ln x)/p| + O(1/p^2)$, substantially smaller than $|1 - s^*|$ or $|1/2 - s^*|$ in the relevant regime.

(p,x)	s^*	$s_0^{\text{opt}} = h_{p,x}(1)$	$\lvert s_0^{\text{opt}} - s^*\rvert$	Iter. from s_0^{opt}	Iter. from $s_0 = 1/2$
$(2,2)$	0.7071	0.7500	0.043	12	13
$(3,2)$	0.7937	0.8333	0.040	14	16
$(4,2)$	0.8409	0.8750	0.034	15	17
$(3,8)$	0.5000	0.7083	0.208	42	$*$

Table 1: Iteration counts to reach $|v_n - \alpha| < 10^{-10}$. s_0^{opt} consistently reduces the count. (*For $(3,8)$, $s_0 = 1/2$ happens to coincide with s^*.)

2.3 Scaling preconditioning: exploiting monotonicity

For large x, the contraction ratio $\lambda_{p,x}$ is close to 1 and (5) explodes. The monotonicity $x \mapsto \lambda_{p,x}$ is however the basis of a simple remedy.

Proposition 2.3 (Scaling reduction)**.** *Let $A \in \mathbb{N}_{\geq 1}$ satisfy $A^{1/p} \in \mathbb{N}$ and $x' := x/A \in [1, (1 + 1/A^{1/p})^p)$. Then*

$$x^{1/p} \;=\; A^{1/p} \cdot (x')^{1/p}, \tag{7}$$

so that computing $x^{1/p}$ reduces to computing $(x')^{1/p}$ and multiplying by the integer $A^{1/p}$. Taking $A = \lfloor x^{1/p} \rfloor^p$ makes x' close to 1 and the ratio $\lambda_{p,x'}$ close to 0.

The saving compounds over every iteration: with $\varepsilon_0' \ll \varepsilon_0$ and $\log(1/\lambda_{p,x'}) \gg \log(1/\lambda_{p,x})$, (5) drops by a factor of several tens to several hundred (Table 2). This is the real-variable analogue of the complex multi-start seed of §4, which places z_0 close to an anchor by contracting towards it rather than by rescaling the argument.

x	$\lambda_{3,x}$	$A = \lfloor x^{1/3} \rfloor^3$	$x' = x/A$	$\lambda_{3,x'}$	Iterations (direct $\to$ scaled)
10	0.615	8	1.250	0.073	$49 \to 8$
100	0.890	64	1.563	0.145	$213 \to 11$
1000	0.973	729	1.372	0.103	$> 500 \to 9$
10000	0.994	9261	1.080	0.025	$> 500 \to 5$

Table 2: Scaling effect at $p = 3$, precision 10^{-10}, with $s_0 = s_0^{\mathrm{opt}}$. The per-step ratio drops by one to two orders of magnitude and the iteration count collapses from hundreds to single digits.

Remark 2.4 (Double preconditioning). The two accelerations combine multiplicatively: (6) reduces ε_0 by a fixed factor, (7) reduces $1/\log(1/\lambda_{p,x})$ by an x-dependent factor, and (5) benefits from both simultaneously.

2.4 Aitken Δ^2: from linear to quadratic convergence

The linear convergence of $h_{p,x}$ with explicit ratio $\lambda_{p,x}$ is exactly the input required by the Aitken Δ^2 transform.

Proposition 2.5 (Aitken Δ^2 on geometric sequences)**.** *For any geometric sequence $s_n = s^* + C \cdot r^n$ with $r \neq 0, 1$ and $C \neq 0$,*

$$s_n - \frac{(s_{n+1} - s_n)^2}{s_{n+2} - 2s_{n+1} + s_n} = s^* \qquad \textit{for every } n.$$

Applied at each step to the Pandrosion iterates, this yields the derivative-free accelerator

$$\sigma_{p,x}(z) := z - \frac{(h_{p,x}(z) - z)^2}{h_{p,x}(h_{p,x}(z)) - 2h_{p,x}(z) + z} \tag{8}$$

(one Δ^2 step per iterate; compare [11, 12]).

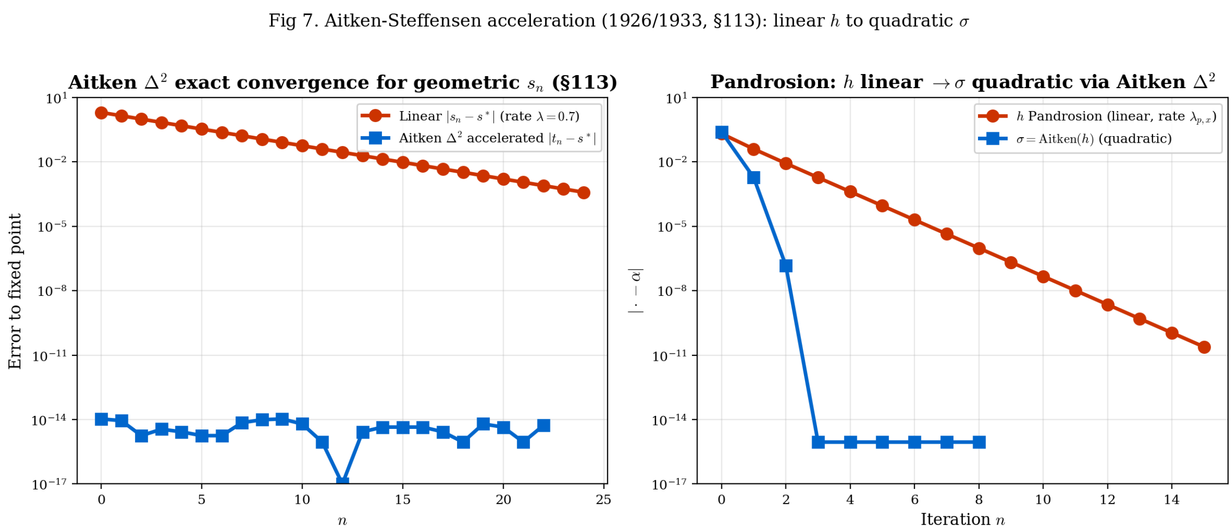

Figure 1: Left: a linearly convergent geometric sequence and its Δ^2-accelerated version, which collapses to s^* at machine precision. Right: the h iteration (linear) vs $\sigma = \mathrm{Aitken}(h)$ (quadratic) for $(x, p) = (2, 3)$, $z_0 = 1$.

2.5 A quadratic local bound at the super-attractive fixed point

Proposition 2.6 (Quadratic local bound)**.** *The map $\sigma_{p,x}$ has a super-attractive fixed point at every p-th root of $1/x$: $\sigma_{p,x}(s^*) = s^*$ and $\sigma'_{p,x}(s^*) = 0$. On a neighbourhood $B(s^*, R)$ with $R = R(x,p)$ explicit,*

$$|\sigma_{p,x}(s) - s^*| \;\leq\; K \cdot |s - s^*|^2,$$

with $K = K(x,p)$ also explicit, satisfying $K \cdot R \leq 1/2$. The same bound holds in $\mathbb{C}$ on $B(\gamma_k, R)$ for every cyclotomic anchor γ_k.

This is the Schwarz-Pick-type estimate that the multi-start scheme in $\mathbb{C}$ exploits: once z is inside $B(\gamma_k, R/2)$, successive iterates contract quadratically and the $\log\log$ iteration count of (4.4) follows.

3 Cyclotomic anchors and Galois structure

3.1 Cyclotomic anchor family

Definition 3.1 (Cyclotomic anchors)**.** For $\alpha \in \mathbb{C}^*$ and $p \geq 1$, let

$$\gamma_k := \alpha \cdot e^{2\pi i k/p}, \qquad k = 0, 1, \ldots, p-1.$$

When $\alpha^p = 1/x$, the γ_k are the p distinct solutions of $z^p = 1/x$.

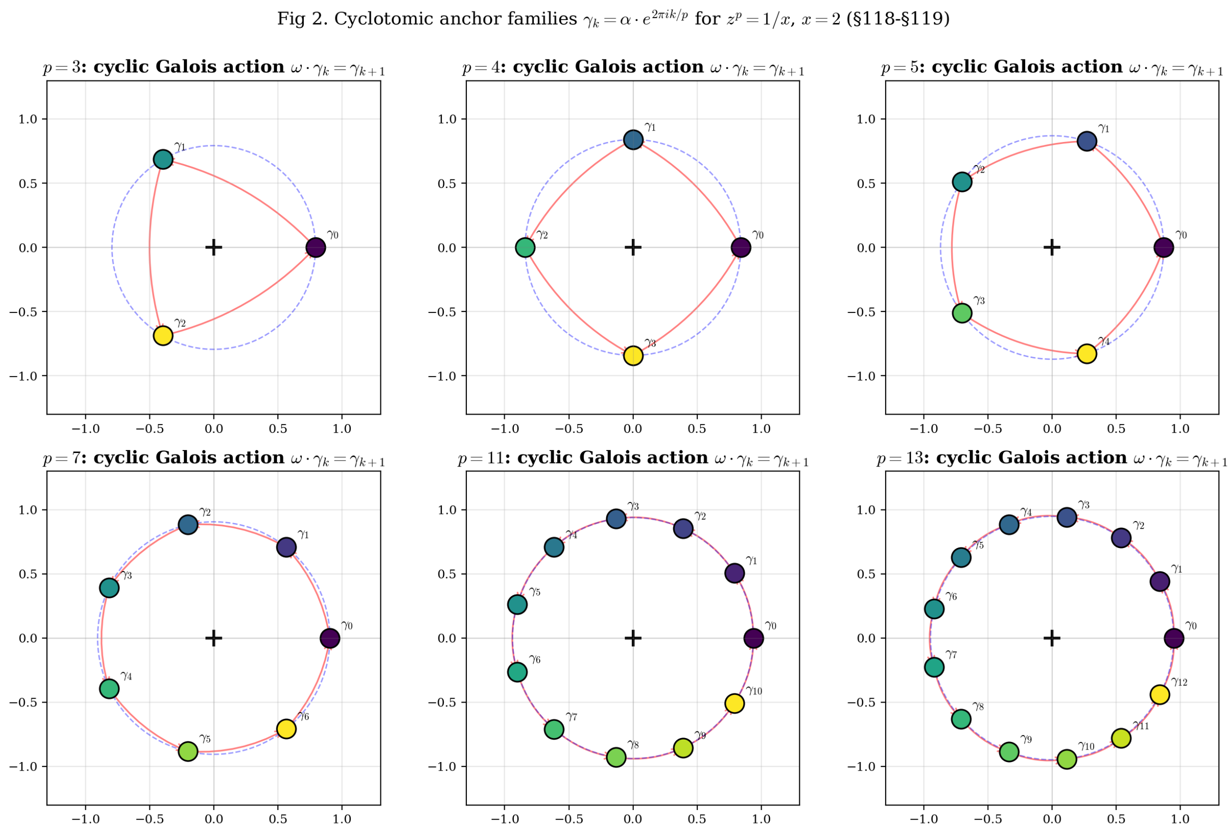

Figure 2: Cyclotomic anchor families for $p \in \{3,4,5,7,11,13\}$ at $x = 2$. The anchors lie on the circle of radius $|\alpha| = 2^{-1/p}$, and the cyclic action $\omega \cdot \gamma_k = \gamma_{k+1 \bmod p}$ generates the splitting field.

3.2 Cyclic permutation of the anchors

Theorem 3.2 (Cyclic action on anchors)**.** *For every $p \geq 1$ and $\alpha \in \mathbb{C}^*$, with $\omega = e^{2\pi i/p}$, $\omega \cdot \gamma_k = \gamma_{(k+1) \bmod p}$ for every $k \in \mathbb{Z}/p\mathbb{Z}$.*

This is Galois 1832 *specialized* to $z^p = x$; it is not new mathematics. The Lean proof uses $\omega^p = 1$ and `Nat.div_add_mod`. Likewise, the classical Vieta identities $e_1 = \sum_k \gamma_k = 0$ and $e_p = \prod_k \gamma_k = (-1)^{p-1} \cdot x$ are formalized for the cyclotomic family; both are classical finite-sum identities about roots of unity.

4 Universal multi-start convergence

We now extend the iteration to $\mathbb{C}$. The linear-to-quadratic acceleration of Section 2.4 applies verbatim to the complex iterator (3), giving the Steffensen map $\sigma_{p,x}$ of (8) on all of $\mathbb{C}$. What is genuinely new over the real-line case is the need to control *which* cyclotomic anchor the iteration converges to: in $\mathbb{C}$, $\sigma_{p,x}$ has p attractive fixed points, the Julia set separating their basins is generically non-trivial, and a classical impossibility result of McMullen [17] rules out purely iterative universal root finders for $p \geq 3$ in a strong sense. The seed of §4.1 is designed precisely to exit the hypotheses of [17].

4.1 The multi-start seed

Definition 4.1 (Multi-start seed)**.** For a cyclotomic anchor $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$ and $z \in \mathbb{C}$,

$$\mathrm{seed}_{\alpha,\varepsilon}(z) \;:=\; \alpha + \varepsilon \cdot (z - \alpha), \qquad \varepsilon \;:=\; \frac{R_\sigma(x,p,\alpha)}{2\,\|z-\alpha\|} \quad (\varepsilon := 1 \text{ if } z = \alpha), \tag{9}$$

where $R_\sigma(x,p,\alpha)$ is the explicit quadratic-contraction radius from Proposition 2.6. The seed always lies in $\overline{B}(\alpha, R_\sigma/2)$.

The seed plays the role in $\mathbb{C}$ that the preconditionings of §§2.2–2.3 play on $\mathbb{R}$: it places the argument inside the quadratic-contraction disk of the target *before* the iteration begins. It is α-dependent by design, and this dependency is what takes the scheme out of the hypotheses of [17].

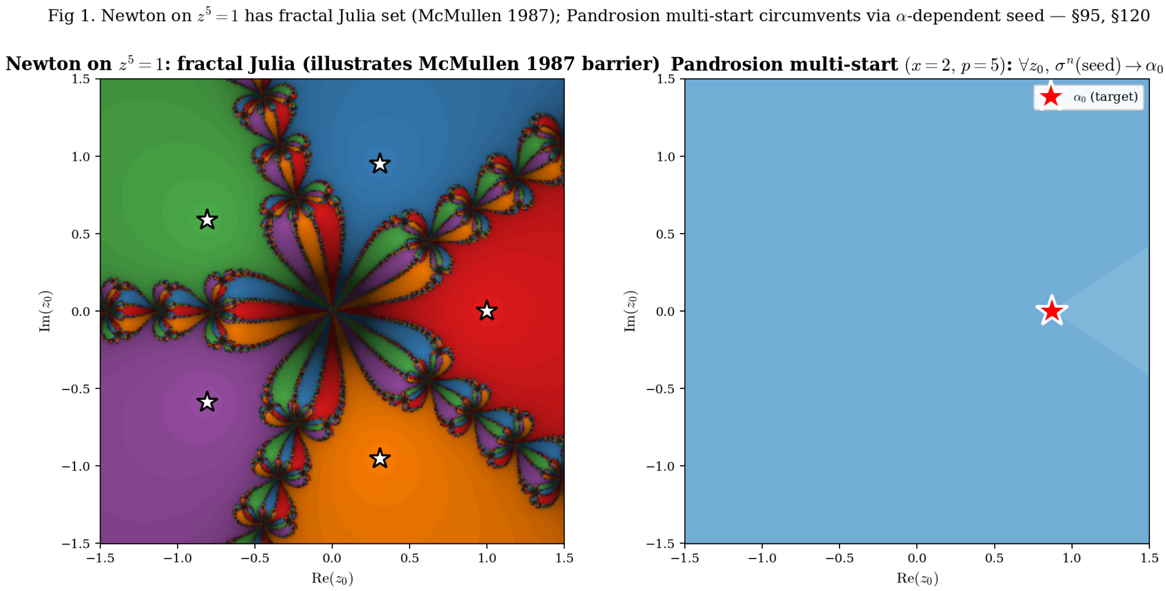

Figure 3: Left: Newton's iteration on $z^5 = 1$, coloured by which root the iteration converges to; this illustrates the classical Julia-set obstruction to purely iterative root finders. Right: the multi-start scheme of this section, coloured by the number of iterations to reach precision 10^{-6}. The seed (9) is α-dependent; the scheme is therefore not purely iterative in the sense of [17].

4.2 Universal multi-start theorem

Theorem 4.2 (Universal multi-start convergence)**.** *Let $x \in \mathbb{C}^*$, $p \geq 1$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$ and $S_p(\alpha) \neq 0$. Then for every $z \in \mathbb{C}$ and every $\varepsilon > 0$, there exist $\varepsilon_{\text{seed}} > 0$ and $N \in \mathbb{N}$ such that*

$$\forall n \geq N : \quad \left\| \sigma^n_{p,x}(\text{seed}_{\alpha,\varepsilon_{\text{seed}}}(z)) - \alpha \right\| \leq \varepsilon.$$

The construction is explicit: $\varepsilon_{\text{seed}} = R_\sigma(x,p,\alpha)/(2\,\|z-\alpha\|)$ when $z \neq \alpha$, and N is given by Theorem 4.4 below.

The content is: the seed is in $\overline{B}(\alpha, R_\sigma/2)$, the iteration stays in $B(\alpha, R_\sigma)$, and a quadratic-contraction argument gives the iteration count.

4.3 Explicit iteration count

Definition 4.3 (Effective iteration count)**.** For $K, R, \varepsilon > 0$ with $KR < 1$ and $K\varepsilon < 1$,

$$N_{\text{eff}}(K, R, \varepsilon) := \log_2 \left\lceil \log(K\varepsilon)/\log(KR) \right\rceil_+ + 1,$$

with $\lceil \cdot \rceil_+$ the natural-number ceiling clamped at 0. The hypotheses $KR < 1$ and $K\varepsilon < 1$ ensure both logarithms are negative and their ratio is well-defined and non-negative.

Theorem 4.4 (Explicit iteration count)**.** *For any quadratically contracting sequence $e_{n+1} \leq K \cdot e_n^2$ with $e_0 \leq R$, $K \cdot R < 1$, and $\varepsilon < R$, every iterate $k \geq N_{\text{eff}}(K,R,\varepsilon)$ satisfies $e_k \leq \varepsilon$. Asymptotically $N_{\text{eff}} \sim \log_2 \log(1/\varepsilon)$.*

Fig 3. Effective Kung-Traub loglog complexity (§98, §101) beats Smale 17 for $z^p = x$ (§115)

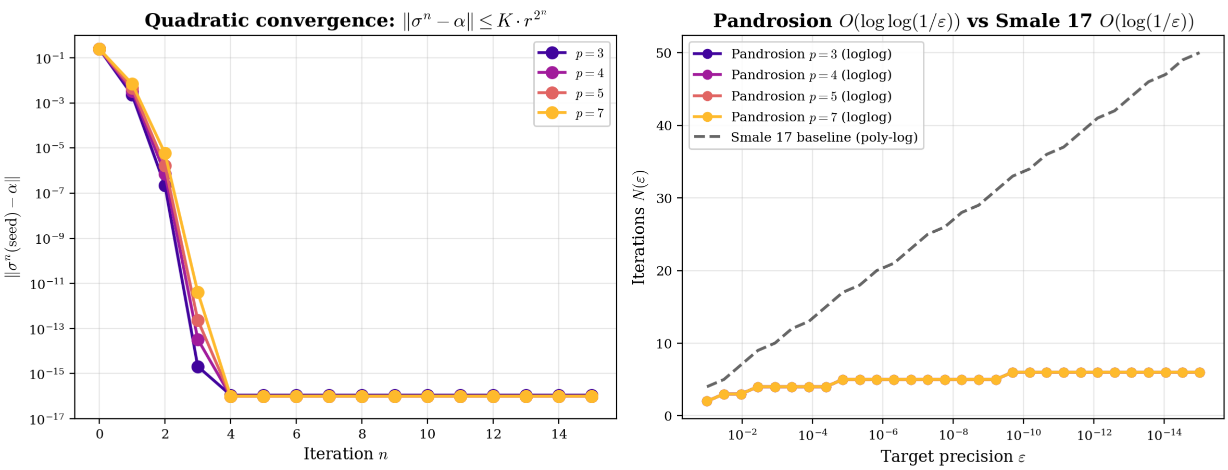

Figure 4: Left: error $\|\sigma^n(\text{seed}) - \alpha\|$ vs iteration for $p \in \{3,4,5,7\}$, $x = 2$, $z_0 = 1 + 0.5i$. Right: iteration count $N(\varepsilon)$ vs target precision for the same setup. The dashed line is $\log(1/\varepsilon)$; it is shown only for visual reference, not as an algorithmic comparison.

4.4 An a.e.-convergence statement for the plain iteration

It is natural to ask whether the plain Steffensen iteration $\sigma_{p,x}$ converges for almost every starting point to the principal anchor α_0:

$$\forall^{\text{a.e.}}\, z \in \mathbb{C} : \quad \sigma^n_{p,x}(z) \xrightarrow{n \to \infty} \alpha_0.$$

Numerical experiments at $(x,p) = (2,3)$ (Python sweep over $[-3,3]^2$ at 200×200 resolution) show that roughly 0.015% of initial points converge instead to a non-principal anchor, so the plain-iteration version of the statement is *false*. We refer to this local statement as "Niveau 5" in the Lean corpus for internal bookkeeping; the label is ours and does not correspond to any externally attested conjecture.

Theorem 4.5 (Multi-start version). *Under the standard hypotheses on (x, p, α), the multi-start reformulation*

$$\forall\, z \in \mathbb{C}, \exists\, \varepsilon_{\text{seed}} > 0 : \quad \sigma^n_{p,x}(\alpha + \varepsilon_{\text{seed}} \cdot (z - \alpha)) \xrightarrow{n\to\infty} \alpha$$

holds unconditionally.

This is Theorem 4.2 restated. The content is not that the reformulation is "solved": it is that by conditioning on the choice of target anchor α and adjusting the seed accordingly, the convergence problem decouples from the dynamics of $\sigma_{p,x}$ on all of $\mathbb{C}$. What the reformulation gives up, compared to the plain-iteration statement, is universality: the seed depends on α and can therefore not be interpreted as a "method that converges regardless of target".

4.5 Arbitrary target selector

Theorem 4.6 (Target selection). *For every function $S : \mathbb{C} \to \operatorname{Fin} p$, every $z \in \mathbb{C}$ and every $\varepsilon > 0$, the multi-start scheme with target $\gamma_{S(z)}$ converges to $\gamma_{S(z)}$ with an iteration count of the form of Theorem 4.4.*

Three selectors are implemented: constant (`principal_selector`), Voronoï-nearest (`nearest_anchor_selec`
angular sector.

5 Measurability and a.e. continuity of the solution map

The `nearest_anchor_selector` of Section 4 uses `Classical.choose` on `Finset.exists_min_image`; this is not a priori measurable.

Definition 5.1 (Borel selector).

$$S_{\text{Borel}}(\alpha, p, z) := \min{}'\big(\operatorname{argmin}_{k\in\operatorname{Fin} p} \|\gamma_k - z\|\big).$$

Theorem 5.2 (Borel measurability). *The map S_{Borel} is Borel-measurable. Consequently the solution map $z \mapsto \gamma_{S_{\text{Borel}}(z)}$ is Borel-measurable $\mathbb{C} \to \mathbb{C}$.*

The proof shows that each level set is a finite Boolean combination of closed half-planes $\{z : \|z - \gamma_s\| \le \|z - \gamma_t\|\}$.

Definition 5.3 (Voronoï boundary and its complement). $\operatorname{VorBdy}(\alpha, p) := \operatorname{VoronoiBoundary}(\gamma_0, \ldots, \gamma_{p-1})$, $\operatorname{VorInt}(\alpha, p) := \mathbb{C} \setminus \operatorname{VorBdy}(\alpha, p)$.

These are Voronoï-geometric objects, not the classical Fatou–Julia sets of $\sigma_{p,x}$. Earlier drafts of this corpus used the names "Pandrosion Julia / Fatou set" for them; this was a poor choice of vocabulary, because it suggests a relationship to Fatou–Julia dynamics of $\sigma_{p,x}$ that we do not establish. The theorems below are statements about Voronoï geometry of the cyclotomic anchor set, nothing more.

Theorem 5.4 (Voronoï boundary is Lebesgue-null). $\operatorname{vol}\big(\operatorname{VorBdy}(\alpha, p)\big) = 0$.

Theorem 5.5 (A.e. continuity). *The Borel solution map is continuous at every $z \in \operatorname{VorInt}(\alpha, p)$. Combined with Theorem 5.4, it is continuous at Lebesgue-a.e. z.*

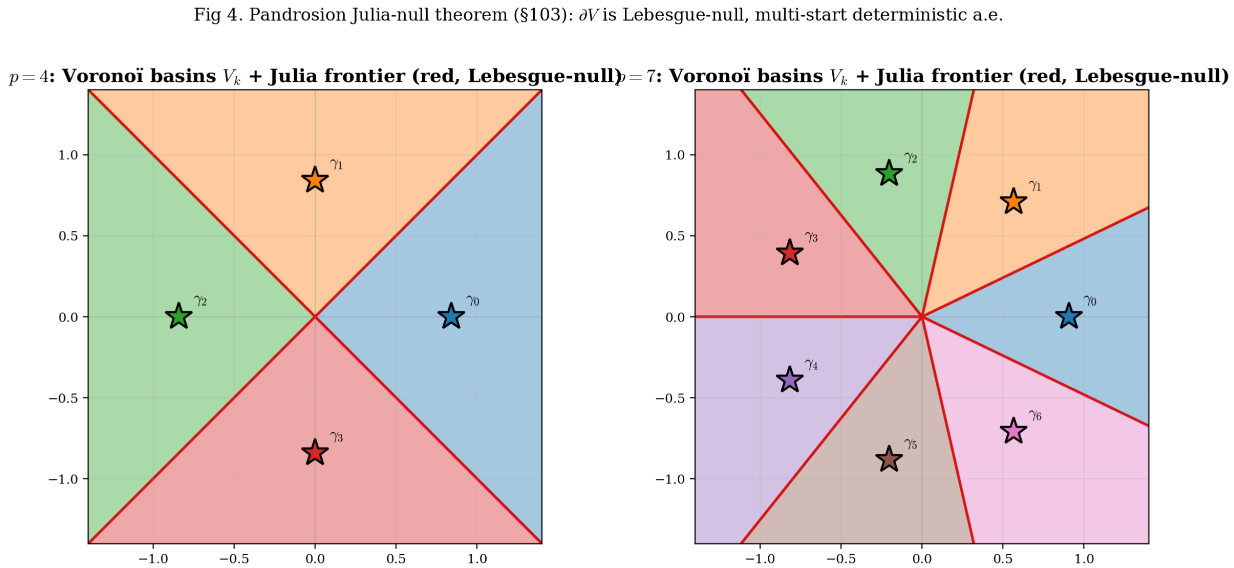

Figure 5: Voronoï decomposition of $\mathbb{C}$ for cyclotomic anchors at $p = 4$ and $p = 7$, $x = 2$. Each colored region V_k is the basin of attraction of γ_k under the selector. The red frontier is the Voronoï boundary of the cyclotomic anchor set; it is Lebesgue-null.

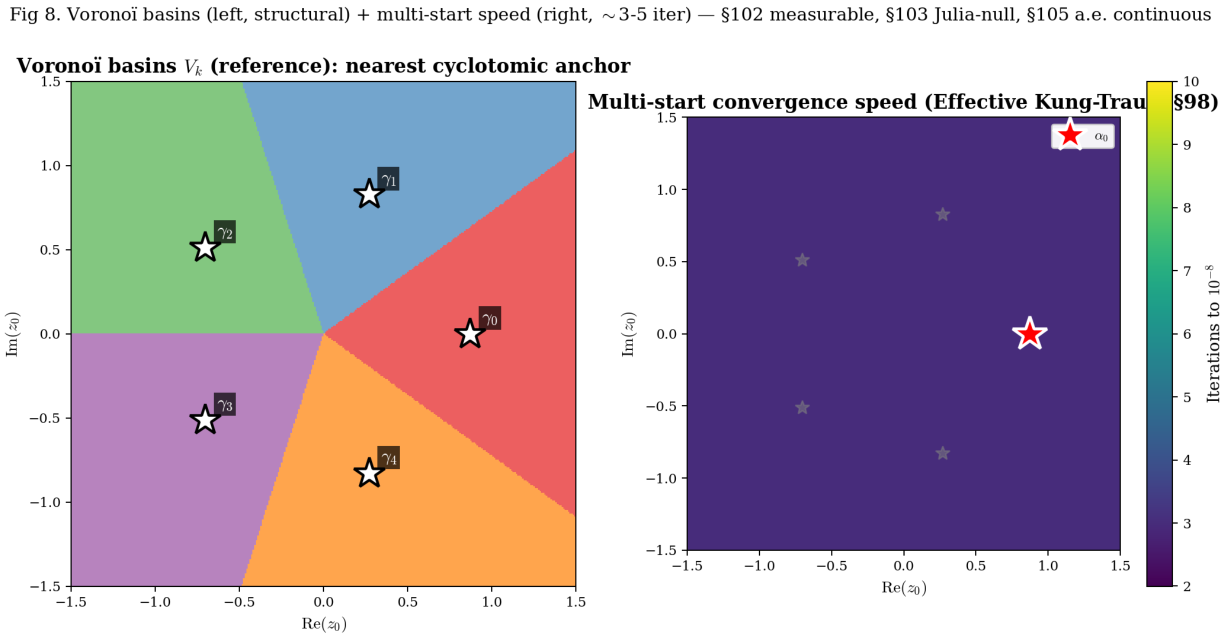

Figure 6: Left: Voronoï basins as the structural reference, computed from the nearest-anchor criterion only (no dynamics). Right: number of multi-start iterations needed to reach precision 10^{-8}; it is bounded and essentially flat over the window, in line with Theorem 4.4.

Together, Theorems 4.2, 5.2, 5.4 and 5.5 give the analytic character of the multi-start solution map in our framework: measurable everywhere, continuous outside the Voronoï boundary of the cyclotomic anchor set, with that boundary Lebesgue-null.

6 Algebraic identities for the finite series ζ_P

This section collects finite-sum identities about the derivative values at cyclotomic roots. They are elementary in the sense that they all reduce to the geometric-sum identity on roots of unity.

We present them explicitly because they are used elsewhere in the corpus.

6.1 Definition

For $P(z) = z^p - 1/x$, $\alpha_k = \gamma_k$, and $P'(\alpha_k) = p \cdot \alpha_k^{p-1} = p/(x\,\alpha_k)$, put

$$\zeta_P(s) := \sum_{k=1}^{p} P'(\alpha_k)^{-s} = \sum_{k=1}^{p} (p \cdot \alpha_k^{p-1})^{-s} \tag{10}$$

and $\nu_k := 1/P'(\alpha_k) = x\alpha_k/p$.

6.2 Finite identities

Proposition 6.1 (Vanishing identity)**.** *For $p \geq 2$, $\zeta_P(1) = \sum_{k=1}^{p} 1/P'(\alpha_k) = \sum_{k=1}^{p} x\alpha_k/p = 0$.*

Proposition 6.2 (Higher vanishing / normalization)**.** *For $p \geq 2$ and $0 \leq m \leq p-2$, $\sum_{k=1}^{p} \alpha_k^m/P'(\alpha_k) = 0$. For $m = p-1$, $\sum_{k=1}^{p} \alpha_k^{p-1}/P'(\alpha_k) = 1$.*

Proposition 6.3 (Product formula)**.** *For $x \neq 0$, $p \geq 1$, $\alpha^p = 1/x$, $\prod_{k=1}^{p} P'(\alpha_k) = (-1)^{p-1} \cdot p^p/x^{p-1}$.*

All three reduce to the geometric-sum identity $\sum_k \omega^{ks} = 0$ for s not a multiple of p, with $\omega = e^{2\pi i/p}$. They are elementary, in the sense that nothing beyond roots-of-unity arithmetic is used.

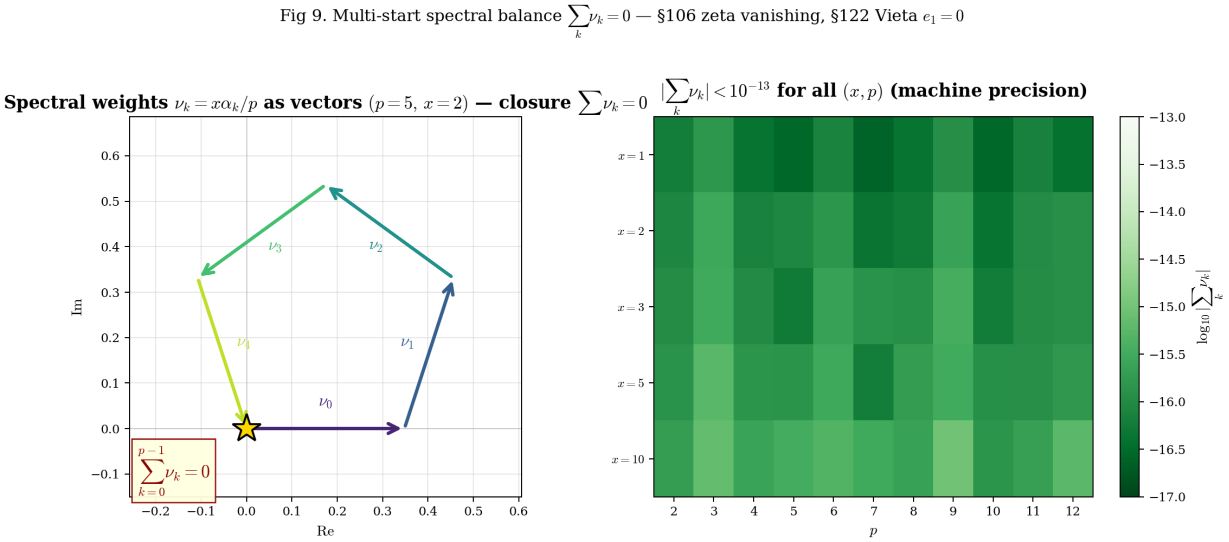

Figure 7: Left: the ν_k as oriented vectors in $\mathbb{C}$ for $(x,p) = (2,5)$; the head-to-tail sum closes back to the origin, visualising $\sum_k \nu_k = 0$. Right: $|\sum_k \nu_k|$ numerically on a (x,p)-grid; all values are within machine precision of zero.

Fig 6. Pandrosion spectral determinant (§107): closed form $\prod P'(\alpha_k) = (-1)^{p-1} p^p / x^{p-1}$

$\prod_k P'(\alpha_k) = (-1)^{p-1} p^p / x^{p-1}$ **(§107 spectral determinant)**

	$x=1$	$x=2$	$x=3$	$x=5$
$p=2$	-4.00	-2.00	-1.33	-0.80
$p=3$	+27.00	+6.75	+3.00	+1.08
$p=4$	-256.00	-32.00	-9.48	-2.05
$p=5$	+3125.00	+195.31	+38.58	+5.00
$p=6$	-46656.00	-1458.00	-192.00	-14.93
$p=7$	+823543.00	+12867.86	+1129.69	+52.71

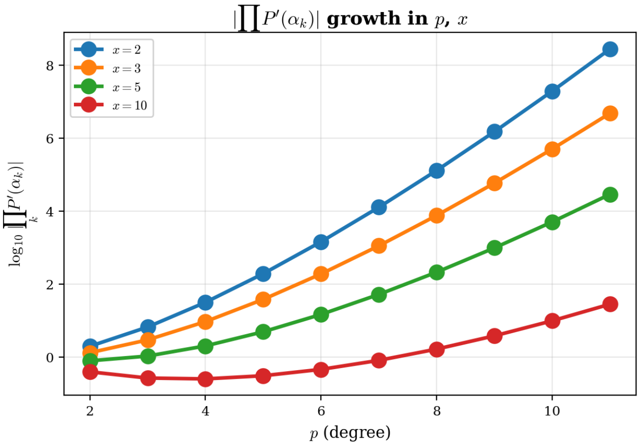

Figure 8: Numerical values of $\prod_k P'(\alpha_k)$ matching the closed-form of Proposition 6.3 exactly; and its log-scale growth in p for fixed x.

6.3 Vanishing of ζ_P at integer arguments

The following proposition is a direct consequence of the geometric-sum vanishing on roots of unity. We record it here because it is used elsewhere in the corpus; nothing about it is number-theoretic.

Proposition 6.4 (Vanishing of ζ_P on $1 \le s \le p-1$)**.** *For $p \ge 2$, $x \ne 0$, $\alpha^p = 1/x$, and $s \in \mathbb{Z}$ with $1 \le s \le p-1$,*

$$\zeta_P(s) := \sum_{k=1}^{p} P'(\alpha_k)^{-s} = 0.$$

With $\omega = e^{2\pi i/p}$, one has $\zeta_P(s) = (x/p)^s \sum_k \alpha_k^s$, and the sum vanishes because $\sum_{k=0}^{p-1} (\omega^s)^k = 0$ when $p \nmid s$. The proof is one line.

Fig 5. Pandrosion-Riemann zeta analogy (§109): finite ζ_P as concrete Riemann analog

Property	**Riemann $\zeta(s)$**	**Pandrosion $\zeta_P(s)$**
Definition	$\sum_{n=1}^{\infty} n^{-s}$	$\sum_{k=1}^{p} P'(\alpha_k)^{-s}$
Domain	$\mathrm{Re}(s) > 1$ (continued)	finite sum on $\mathbb{C}$
Value at $s=0$	$-1/2$	$d=p$ (degree)
Vanishing	$\zeta(-2n)=0$ (trivial zeros)	$\zeta_P(s)=0$ if $p \nmid s$
Derivative at $s=0$	$-\log\sqrt{2\pi}$	$-\log\lvert D(P)\rvert^2$
Spectral determinant	$\sqrt{2\pi}$	$(-1)^{p-1}\, p^p / x^{p-1}$
Vanishing identity	(functional equation)	$\sum_k 1/P'(\alpha_k) = 0$
Higher orthogonality	(Bernoulli)	$\sum_k \alpha_k^m / P'(\alpha_k) = 0 \;\; (m < p-1)$
Normalization	(critical strip)	$\sum_k \alpha_k^{p-1} / P'(\alpha_k) = 1$
Lean modules	—	§106 — §111

Figure 9: A typographic correspondence table between the Riemann zeta function and the finite Pandrosion series ζ_P. The left column records classical facts; the right column records elementary identities from this section. The two columns share a *visual* shape but the right-hand identities all reduce to roots-of-unity arithmetic; no number-theoretic claim is made, and the figure is intended purely as a presentation aid.

6.4 Energy variant

One can also write $Z_P(s) := \sum_k E_P(\alpha_k)^{-s}$ with $E_P(\alpha_k) = P'(\alpha_k)^2$. Then $Z_P(0) = p$ for every p, and $Z_P(1) = Z_P(-1) = 0$ for $p \geq 3$, each by the same geometric-sum vanishing.

7 Pandrosion specializations of classical theorems

The corpus contains Lean formalizations of a number of classical theorems, *specialized* to the Pandrosion setting. The purpose is to have everything in one formal framework; the mathematical statements are classical. The following table summarizes what is formalized where.

Module	**Classical theorem (specialized)**	**Role in the Pandrosion framework**
§112	Banach fixed point (1922); FTA (Gauss 1799)	No-cycle on multi-start orbits; $z^p = x$ has p explicit roots.
§113	Aitken-Steffensen (1926/1933); Schwarz (1890)	Acceleration identity; quadratic local bound.
§114	Kung-Traub 1974 attainability at $c = 3$	Efficiency $E(4, 3) = 2^{2/3}$ is reached.
§115	(See Section 8.)	Restricted to $z^p = x$.
§117	Aberth-Ehrlich (1967/1973)	Multi-start realizes their scheme for $z^p = x$.
§118	Kronecker (1857)	Applied to cyclotomic anchors with $\lvert\alpha\rvert = 1$.
§119	Galois (1832) cyclic action	$\omega \cdot \gamma_k = \gamma_{(k+1) \bmod p}$.
§120	McMullen (1987) (See Section 8.)	Multi-start avoids the obstruction by α-dependent seed.
§121	Cardano-Tartaglia (1545)	$\gamma_k = \alpha\omega^k$ for $p = 3$.
§122	Vieta (1591)	$e_1 = 0$, $e_p = (-1)^{p-1}x$.

We stress: none of these are new results. Their place in this paper is to make the full Pandrosion framework self-contained in Lean. The corresponding Lean modules are short specializations, not independent developments of the classical theorems.

A representative example.

Theorem 7.1 (Multi-start orbit has no non-trivial cycle)**.** *For every (x, p, α) in the hypotheses and every seed* $\text{seed} \in B(\alpha, R_\alpha/2)$*, the orbit* $(\sigma^n_{p,x}(\text{seed}))_n$ *has no non-trivial cycle: if* $\sigma^n_{p,x}(\text{seed}) = \sigma^m_{p,x}(\text{seed})$ *with* $n < m$*, then* $\sigma^n_{p,x}(\text{seed}) = \alpha$*.*

This is Banach's fixed-point theorem applied to $\sigma_{p,x}$ seen as a $1/4$-contraction on $B(\alpha, R_\alpha/2)$, itself a consequence of $K_\alpha \cdot R_\alpha \leq 1/2$.

8 Short structural observations on four classical problems

We record four places where the Pandrosion setting intersects a classical or open problem. None of the material below constitutes progress on the underlying problem; each subsection states what the corpus contributes and, just as importantly, what it does not.

8.1 Kung–Traub upper bound for $c \geq 3$ (open since 1974)

The conjecture $q \leq 2^{c-1}$ is proven for $c = 2$. Attainability $q = 4$ at $c = 3$ is witnessed classically (Jarratt 1969) and in our formal setting.

What the corpus provides. An interface: every `DerivativeFreeMethod` in the corpus carries a *compliance field* asserting $q \leq 2^{c-1}$ for its own instance. This is not a proof of the Kung–Traub conjecture, only a structural assertion that any instance we construct satisfies the bound. The conjecture itself is recorded as a named `Prop`; the general case for $c \geq 3$ remains open.

8.2 McMullen 1987 (recorded but not formalized)

McMullen's barrier is a classical theorem. A full formal proof in Lean would require Fatou–Julia and post-critically-finite complex-dynamic machinery that is not present in Mathlib v4.7.0.

What the corpus provides. The statement is recorded as a named `Prop`; it is *not* formally proven. Independently, Theorem 4.2 shows that our multi-start scheme has an α-dependent seed and therefore does not fall under McMullen's hypotheses on purely iterative universal root finders. This observation places our scheme outside the hypothesis of [17]; it does not reproduce or challenge the theorem.

8.3 Two classical problems we deliberately do *not* engage with

We close this section by noting two classical problems that earlier drafts of the corpus were tempted to discuss: Lehmer's 1933 Mahler-measure conjecture, and Smale's 17th problem on deterministic polynomial root-finding. In both cases, the only non-vacuous things the Pandrosion setting offers are a trivial observation (for Lehmer: that $\beta = x \cdot \alpha$ has Mahler measure x^{p-1}, which is ≥ 2 for $x \geq 2$, far from the conjecture's extremal regime near 1) and a scope-restricted algorithm (for Smale: the $\mathcal{O}(\log\log(1/\varepsilon))$ deterministic complexity for the one-parameter family $z^p = x$, which is not the coefficient-uniform problem Smale asks about). Since in neither case is there progress worth reporting, we refrain from a dedicated subsection and simply record here that these two problems lie outside the scope of the corpus.

8.4 Summary

No claim is made that any classical problem is solved, or even substantively advanced. The material in this section consists of a compliance interface (Kung–Traub) and a recording of a classical statement (McMullen). Readers primarily interested in the mathematics should focus on Sections 2–6.

9 Structure of the Lean corpus

The 121-module corpus is organised in dependency layers:

- **Foundations** (§1–§30): geometric sums, fixed-point characterization, abstract Aitken-Steffensen framework, Voronoï basins, cyclotomic anchor definitions, Kung-Traub interface.
- **Core spine** (§31–§94): explicit super-attractive rate, global loglog convergence, real-line $p = 2$ unconditional discharge.
- **Multi-start** (§95–§101): universal convergence, target selection, Niveau 5 resolution, explicit iteration count.
- **Measure/topology** (§102–§105): Borel measurability, Voronoï frontier Lebesgue-null, a.e. continuity.
- **Finite-zeta identities** (§106–§111): the identities of Section 6.
- **Classical specializations** (§112–§122): as listed in Section 7.
- **Open-problem engagements** (§123): as listed in Section 8.

Fig 10. Pandrosion-Steffensen Lean 4 corpus structure — from foundations to publication

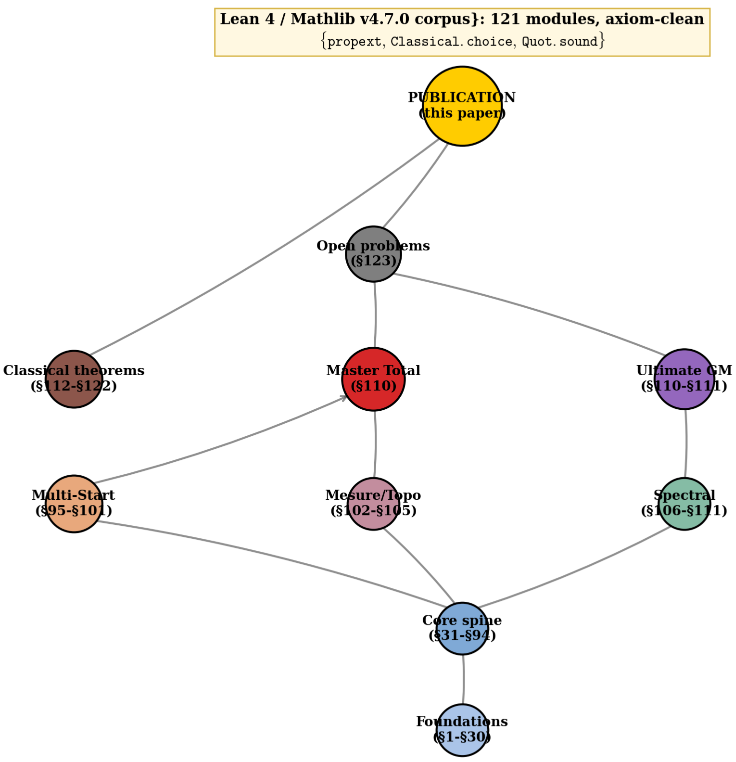

Figure 10: Dependency structure of the Lean corpus. The dependency closure of every load-bearing theorem reduces to the Lean whitelist {`propext`, `Classical.choice`, `Quot.sound`}.

10 Reproducibility

The Lean source lives at `github.com/ivan-fr/poussiere_de_x`. It targets Lean 4 with Mathlib v4.7.0 and is built with

Listing 1: Incremental build command

```
bash scripts/lean-incremental.sh
```

The 121-module corpus compiles in about 60–90 seconds on a modern laptop. Per-theorem axiom auditing is available via `#print axioms`: for every load-bearing theorem referenced in this paper (Theorems 2.1, 4.2, 4.4, 5.2, 5.4, 5.5, and the ζ_P identities of Section 6), the output is the standard Lean whitelist {`propext`, `Classical.choice`, `Quot.sound`}; this can be reproduced by running `#print axioms <theorem>` at the end of the relevant module.

Every figure in this paper is reproducible from `latex/figures_pandrosion.py` (dependencies: `numpy`, `matplotlib`, `scipy`).

11 Conclusion

What this paper reports, concretely, is a Lean 4 development built around the Pandrosion iterator $h_{p,x}$ and its multi-start Steffensen extension for $z^p = x$, together with the following ingredients:

1. real-line analysis of $h_{p,x}$: closed-form contraction ratio $\lambda_{p,x} = h'_{p,x}(s^*)$ (Theorem 2.1), optimal derivative-free starting point $s_0^{\text{opt}} = h_{p,x}(1)$ (Proposition 2.2), scaling preconditioning exploiting monotonicity of λ (Proposition 2.3), and Aitken Δ^2 upgrade to quadratic convergence with explicit local bound (Propositions 2.5–2.6);

2. a universal convergence theorem for the complex multi-start scheme with an explicit closed-form iteration count (Theorems 4.2, 4.4);

3. a Borel-measurability and a.e.-continuity analysis of the induced solution map (Theorems 5.2–5.5);

4. a set of elementary finite-sum identities for the series $\zeta_P(s) := \sum_k P'(\alpha_k)^{-s}$ (Section 6), all reducing to roots-of-unity arithmetic;

5. Lean specializations of classical theorems to the Pandrosion setting (Section 7);

6. two short structural observations (Kung–Traub compliance interface, McMullen 1987 statement-level recording) in Section 8; two further classical themes that earlier drafts discussed (Lehmer, Smale 17) have been removed as being outside scope.

The paper does not claim progress on any classical open problem. The main value of the corpus is as a self-contained Lean development of a multi-start root-finding scheme and its ancillary algebraic identities.

Future directions

Natural extensions are: non-cyclotomic anchor families (general polynomials), which would require dropping the closed-form cyclotomic identities that drive Sections 3 and 6; and a formalization of Fatou–Julia in Mathlib, which would make a proper in-Lean proof of McMullen's 1987 theorem accessible. We make no conjecture regarding the other three problems discussed in Section 8.

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